

Underground intelligence: how roots explore their world

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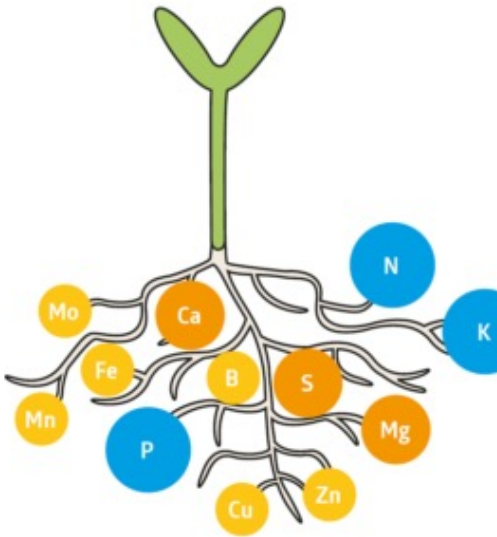


The hidden half of plants

- 22% of total plant biomass in forests
- 67% in grasslands
- They anchor the plant
- Absorb water and nutrients
- Interact with microbes
- and respond to changes in the soil

Plants cannot move away, they must
adapt in place



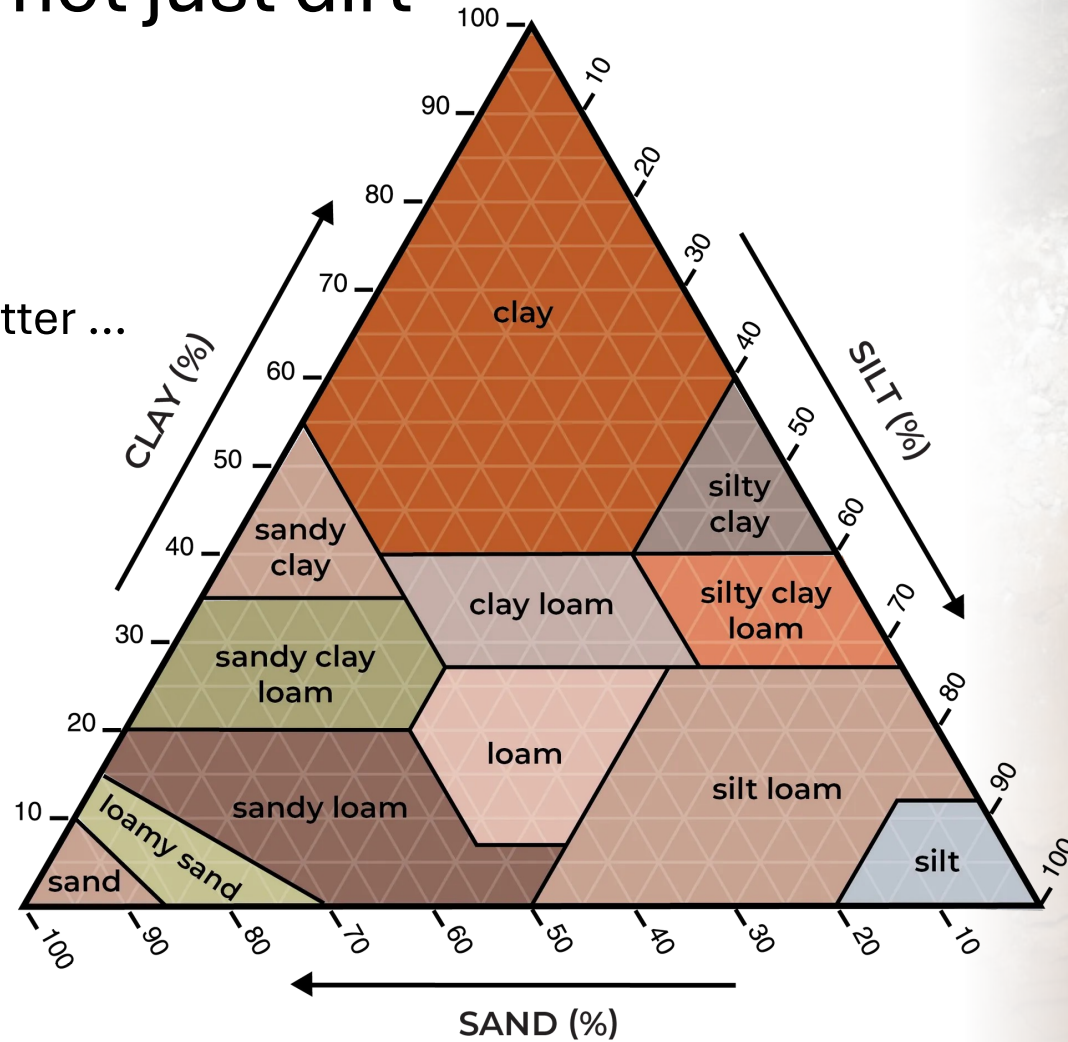


And most plant challenges start below ground

- Limitation of water
- Nutrients unevenly distributed
- Salt and toxic compounds
- Absorb water and nutrients
- Pathogens and microbial communities

The soil is not just dirt

- Texture
- Chemistry
- organic matter ...





An underground city:

Roots are not just pipes. They sense, grow, decide and adapt

- **Sensors** → cells detect soil signals
- **Gates** → barriers control entry
- **Roads** → vascular tissues transport resources
- **Workers** → specialized cells build and remodel tissues

water → where to grow

nutrients → where resources are

salt / stress → danger

signal hormones → internal instructions

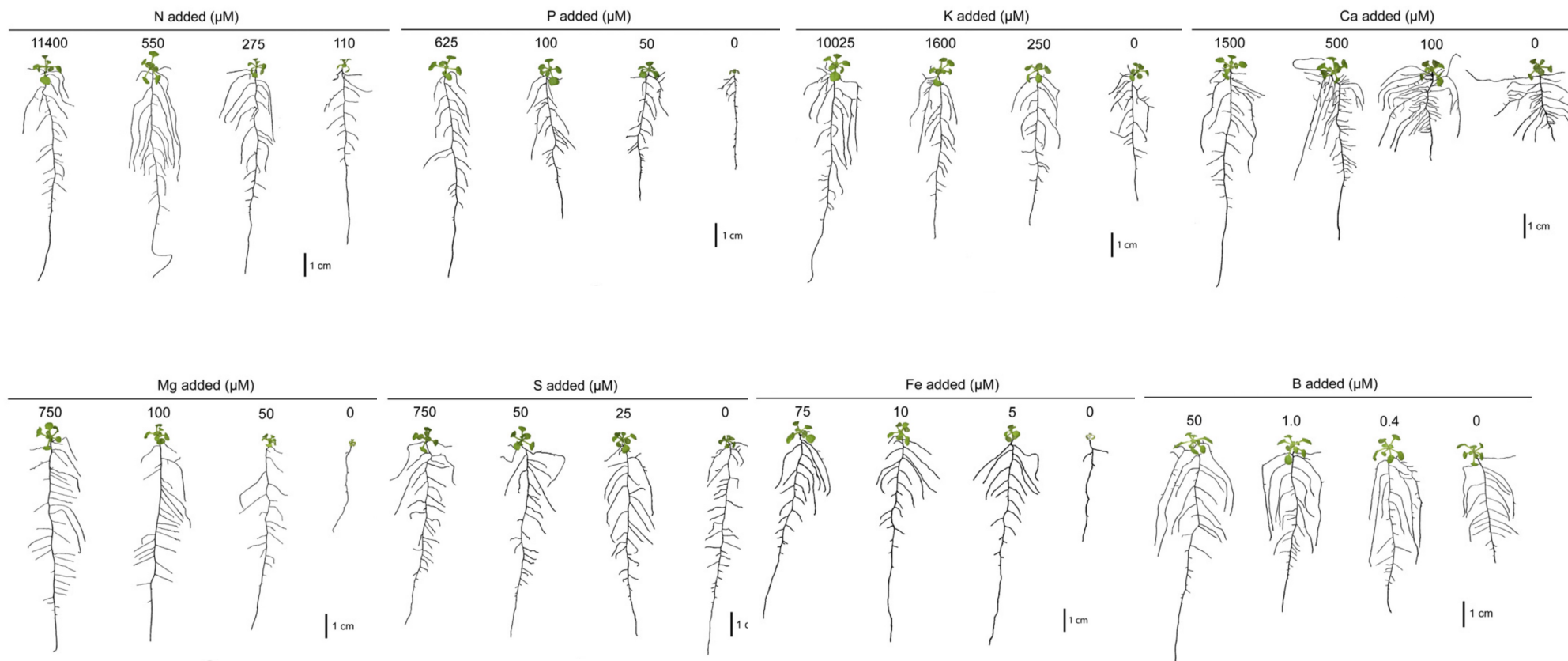
microbes → friends or enemies

gravity / light → orientation cues

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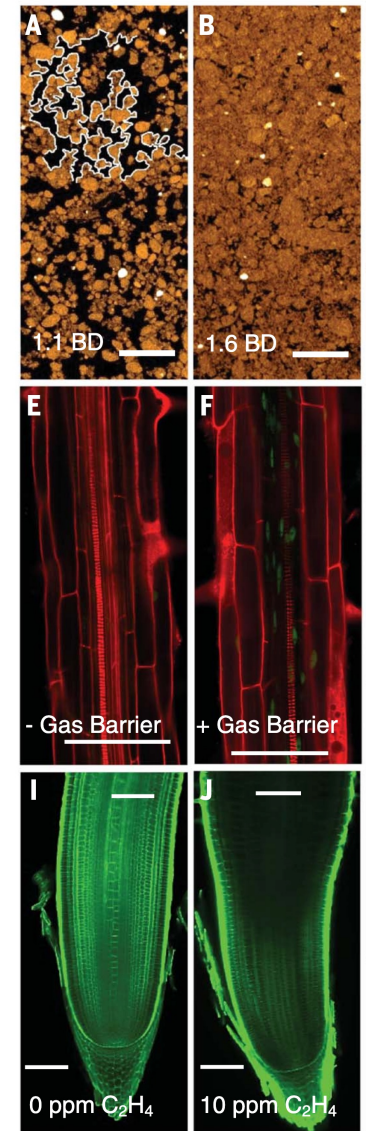
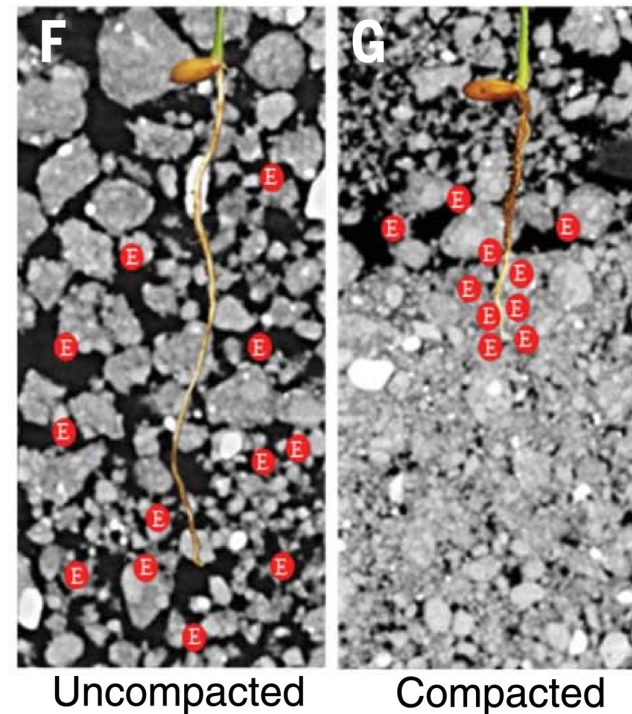
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Soil compaction (ex: through restricted ethylene diffusion)

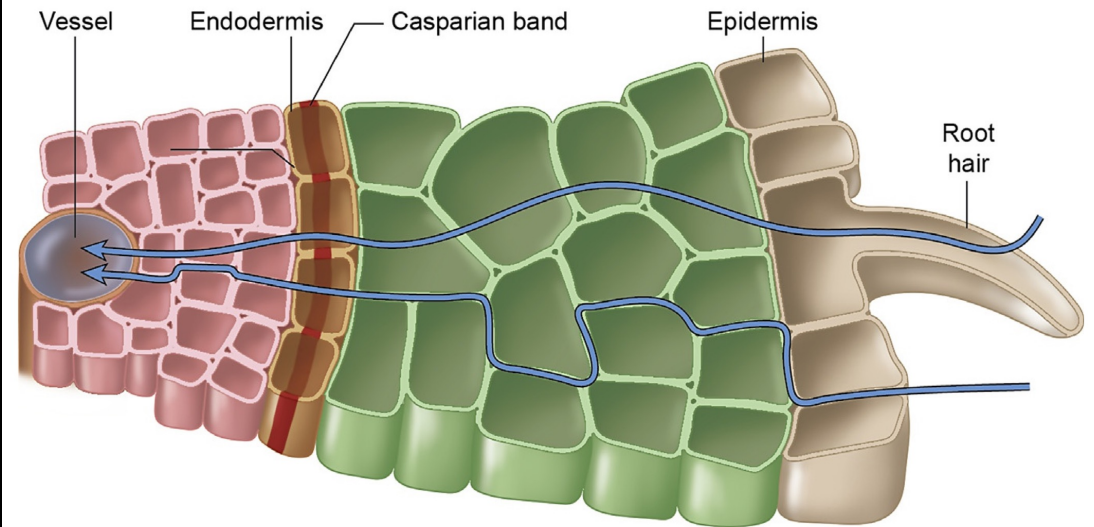
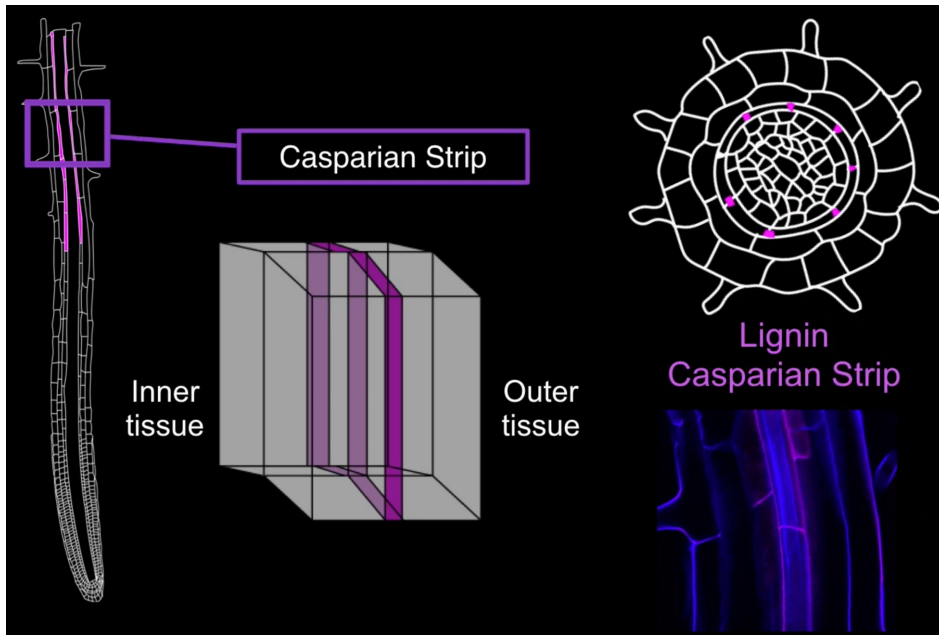
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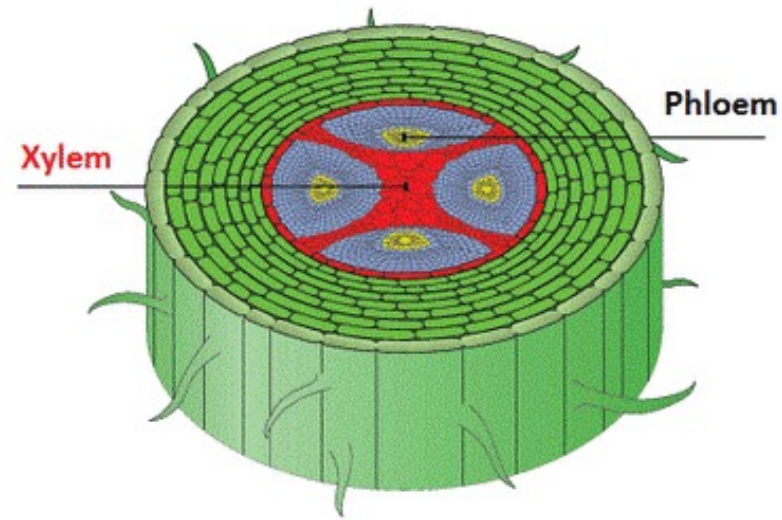
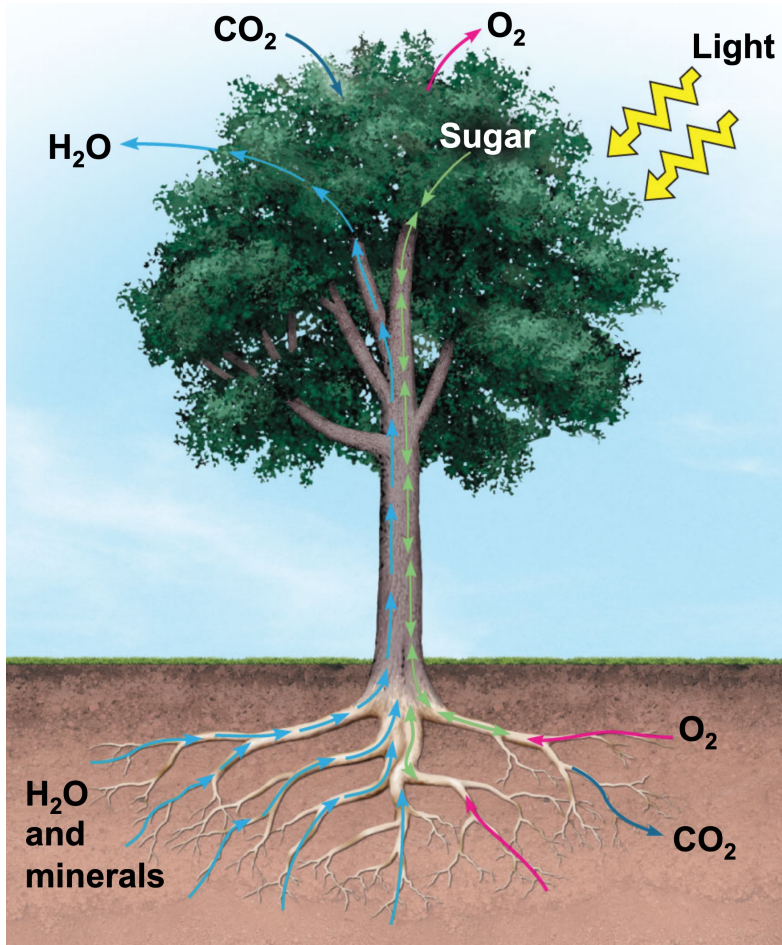
(Pandey et al., 2021)



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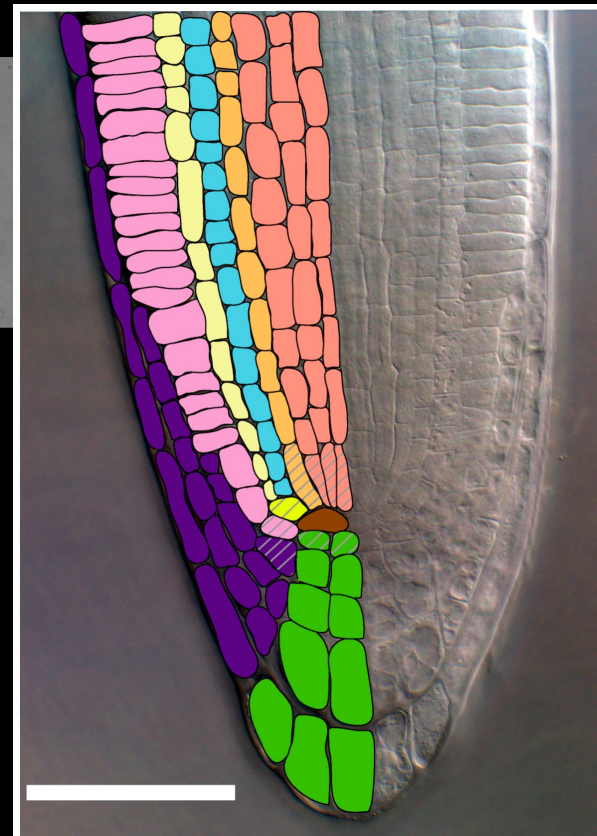


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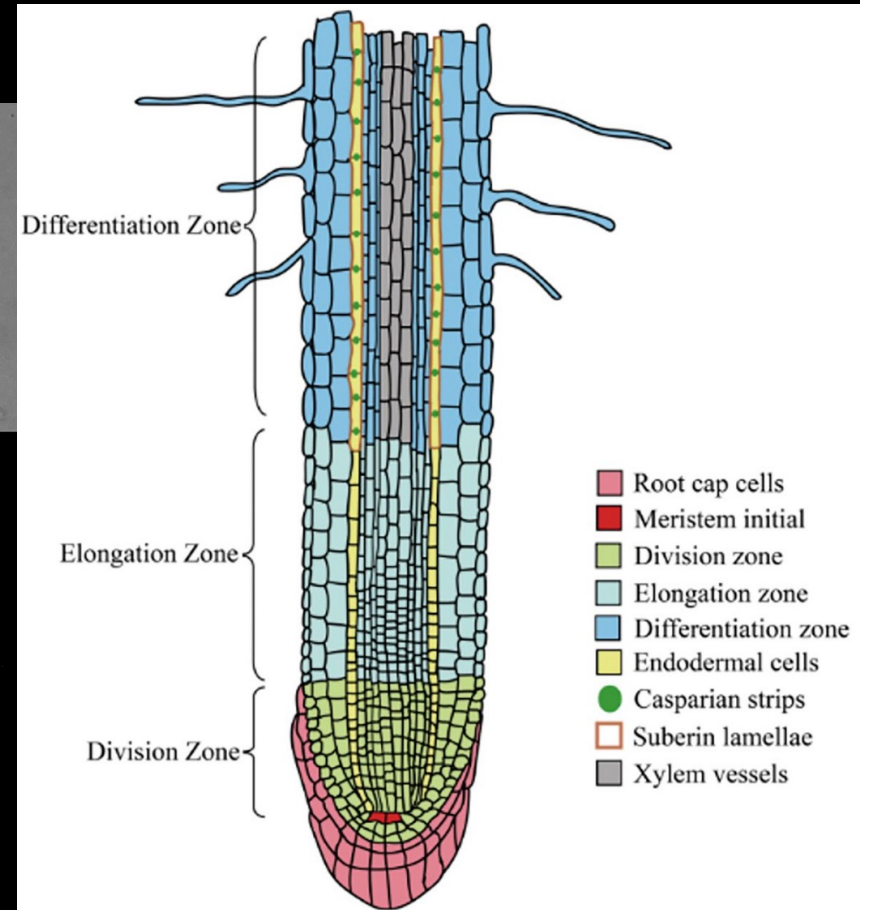
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One root, many tissues

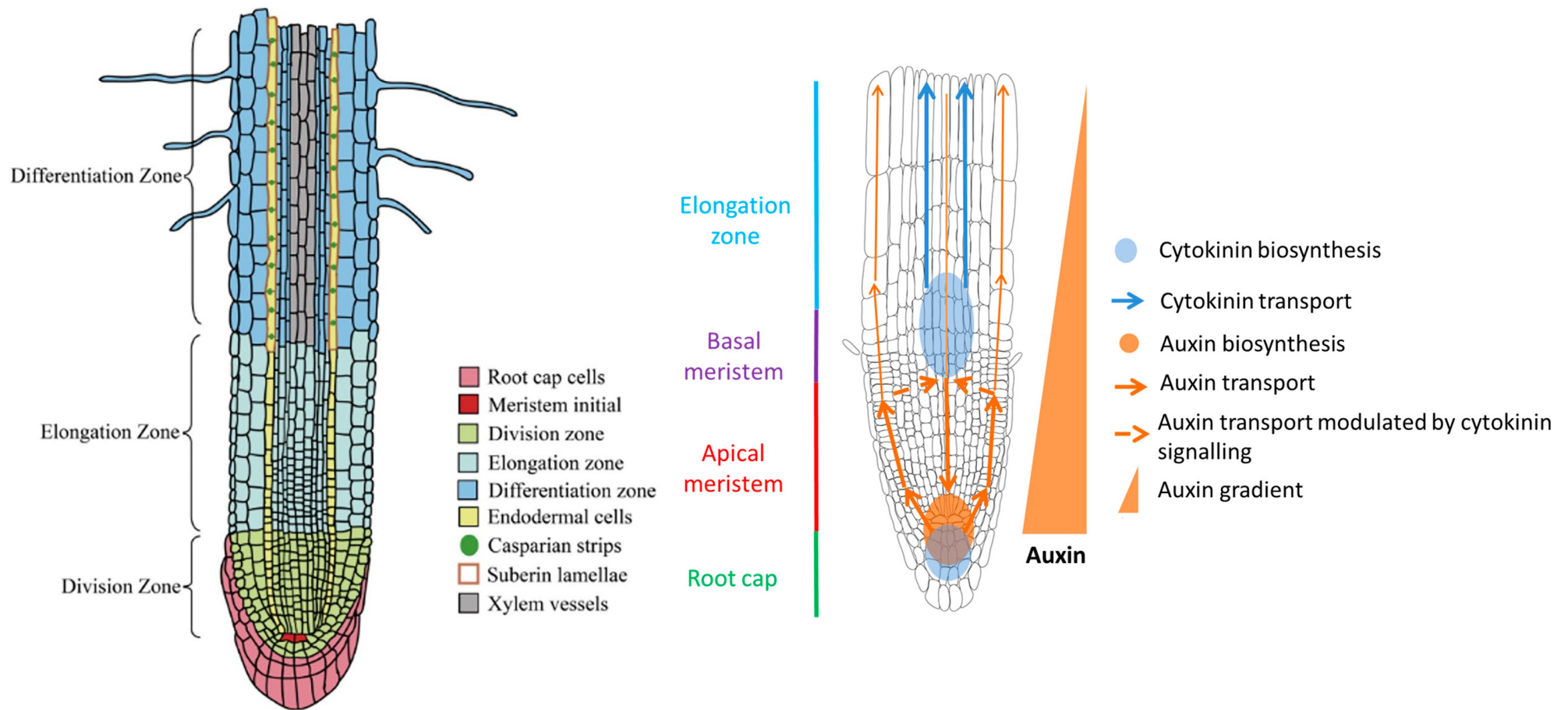


- Epidermis
- Cortex
- Endodermis
- Pericycle
- Vascular tissues
- Quiescent Center
- Columella
- Lateral root cap
- Cortex endodermis initial
- Initials

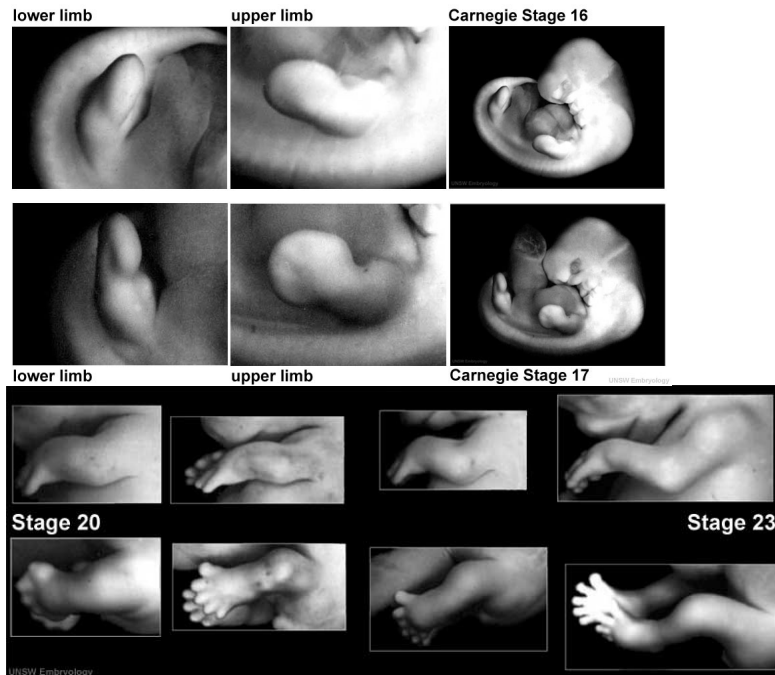
How does a root grow ?



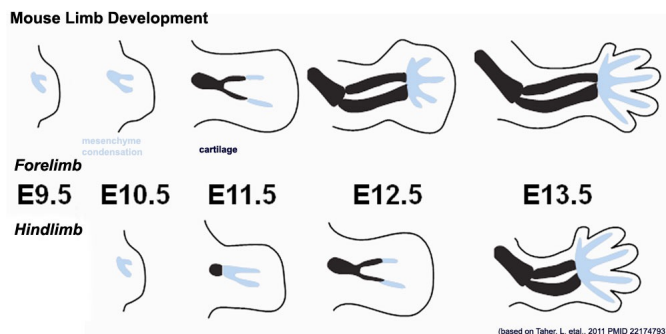
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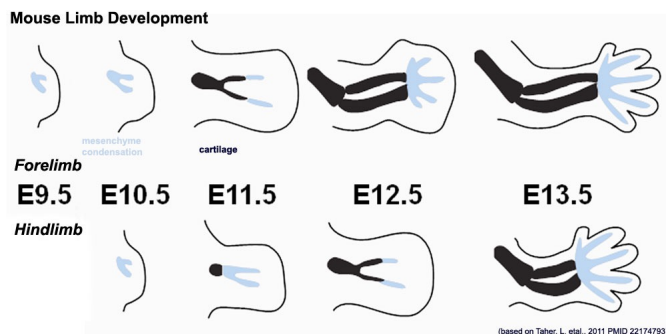
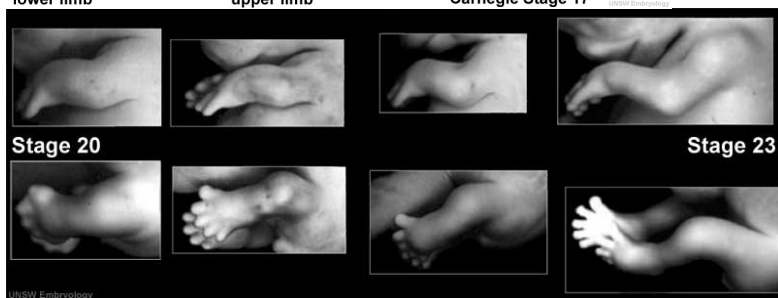
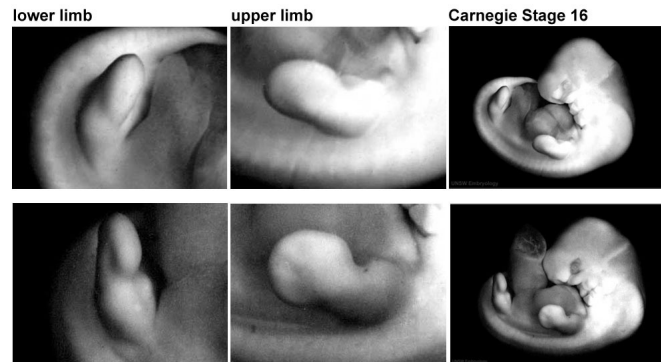
How does a new root emerge?



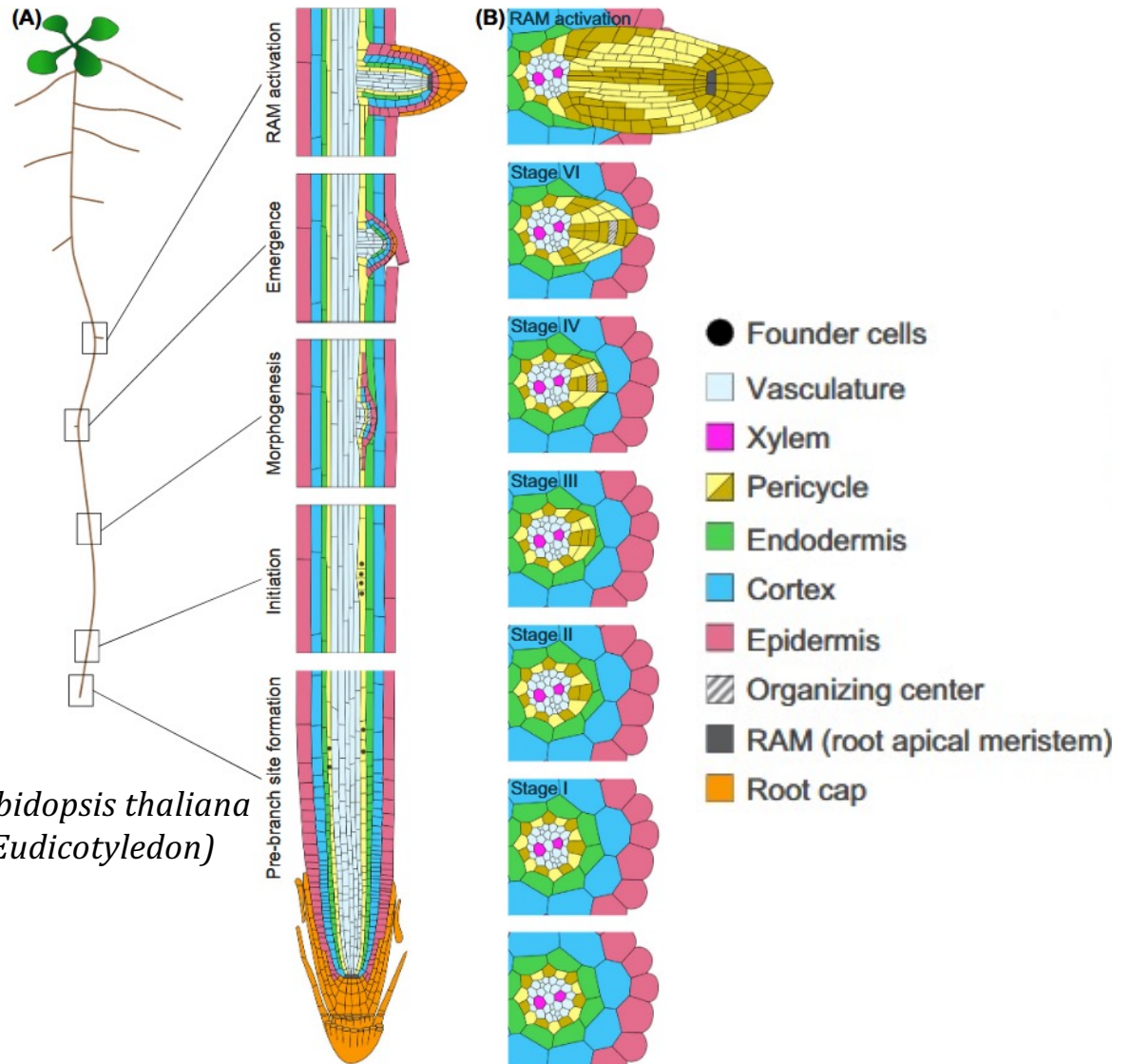
Animals are born with most of their organs already formed. Plants, instead, keep building new organs throughout life.

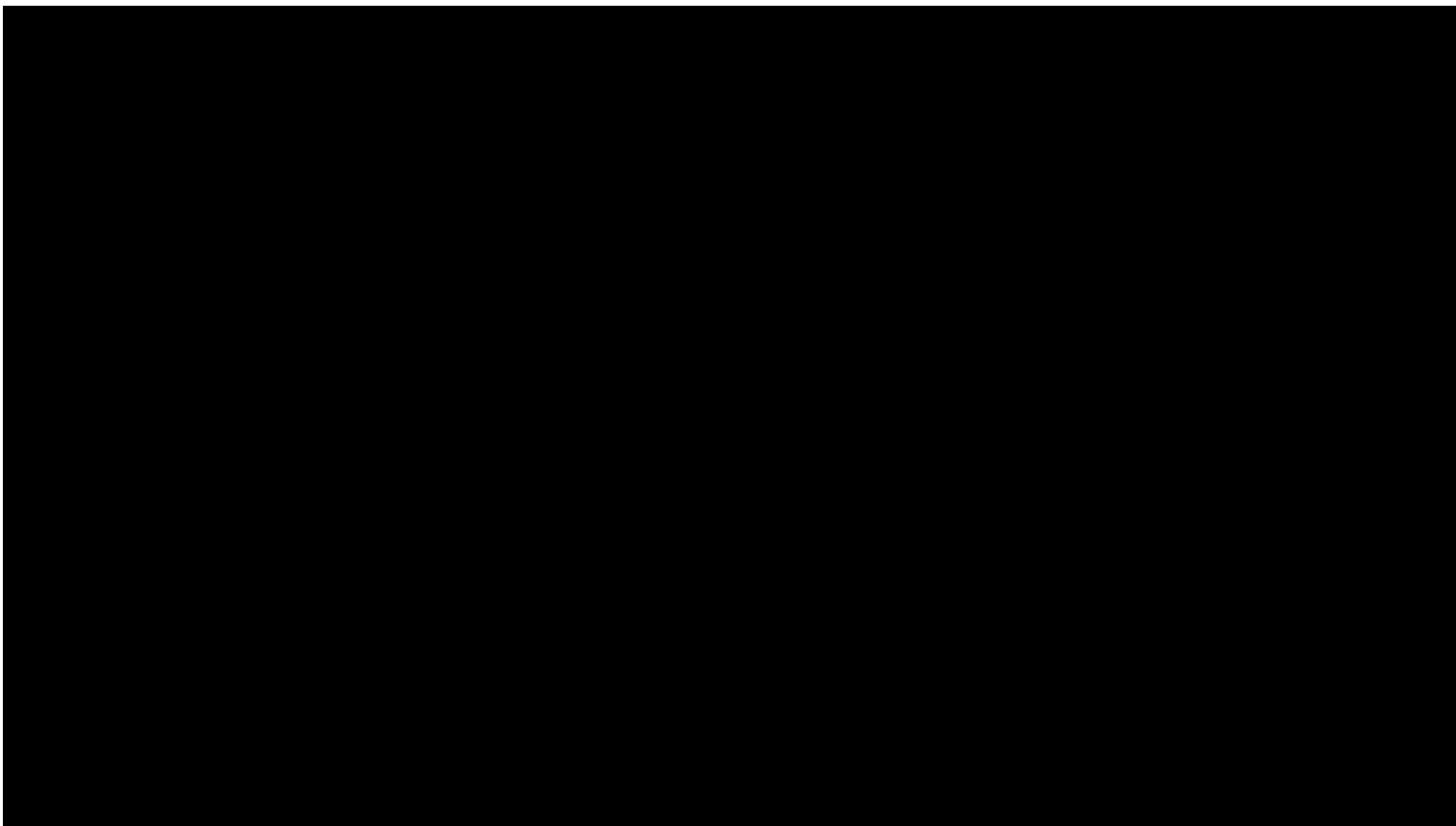


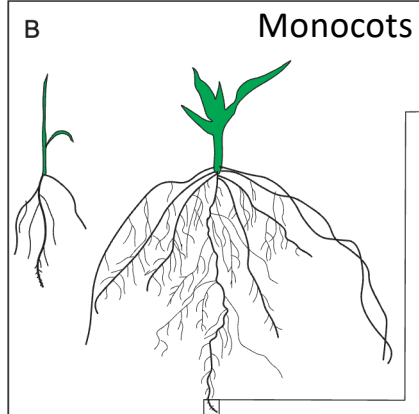
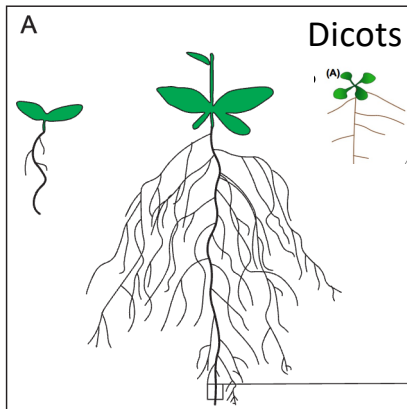
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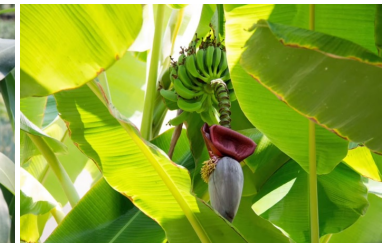
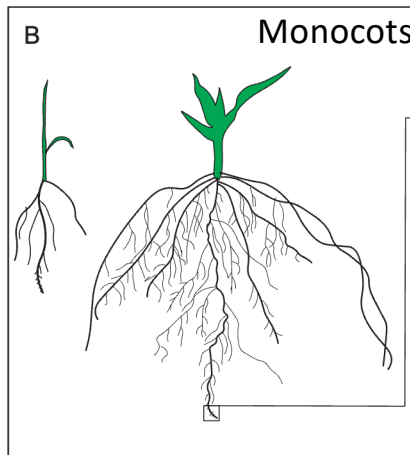
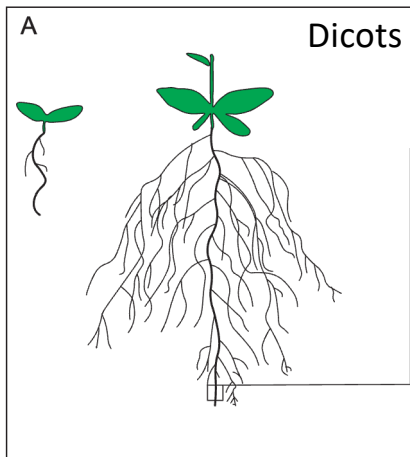


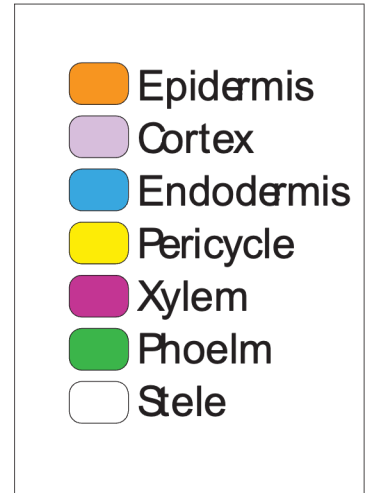
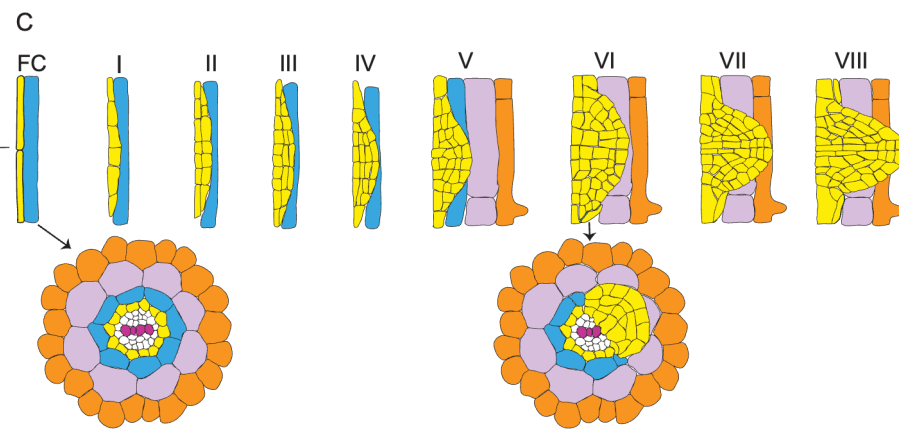
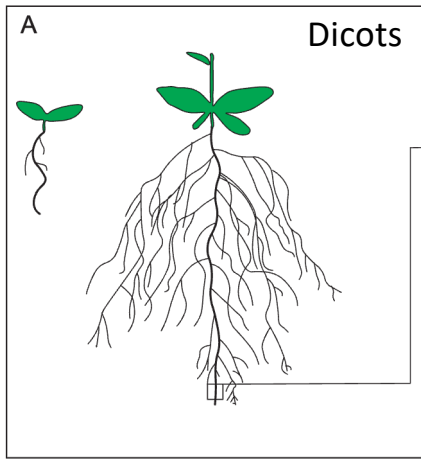
Arabidopsis thaliana
(Eudicotyledon)

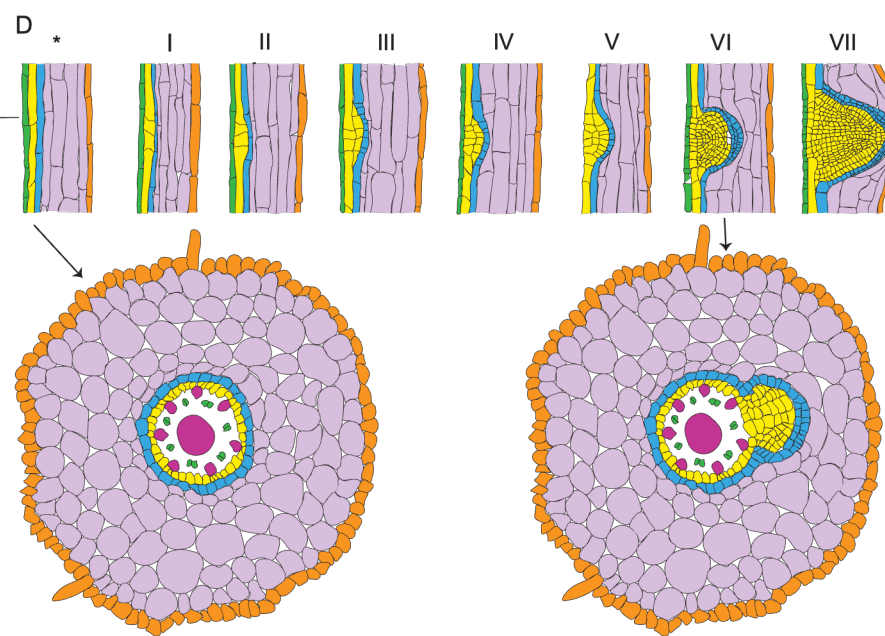
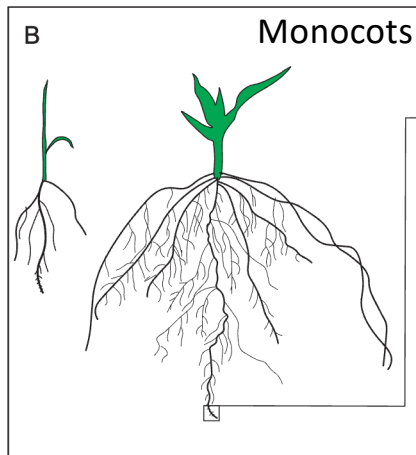
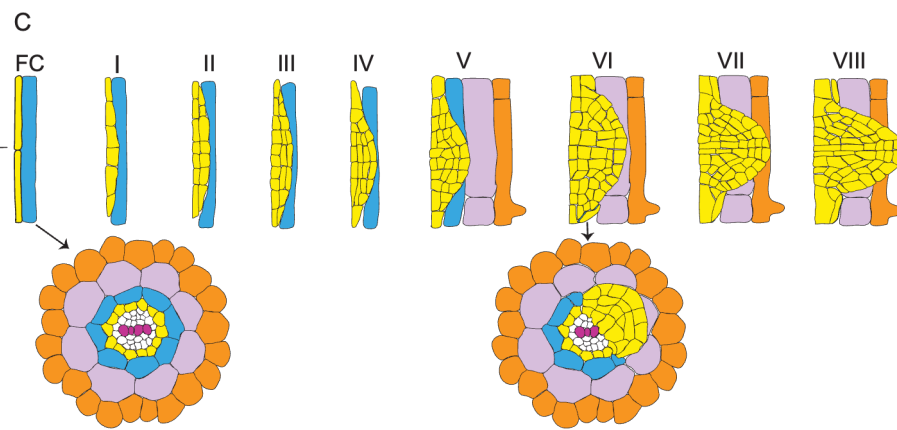
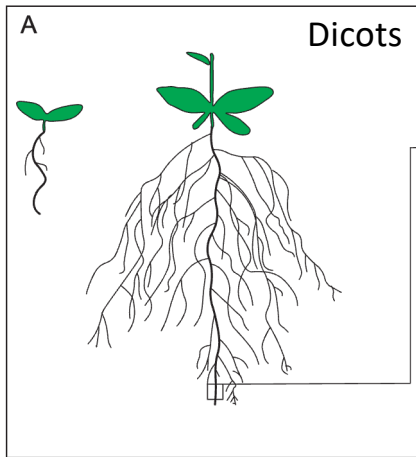






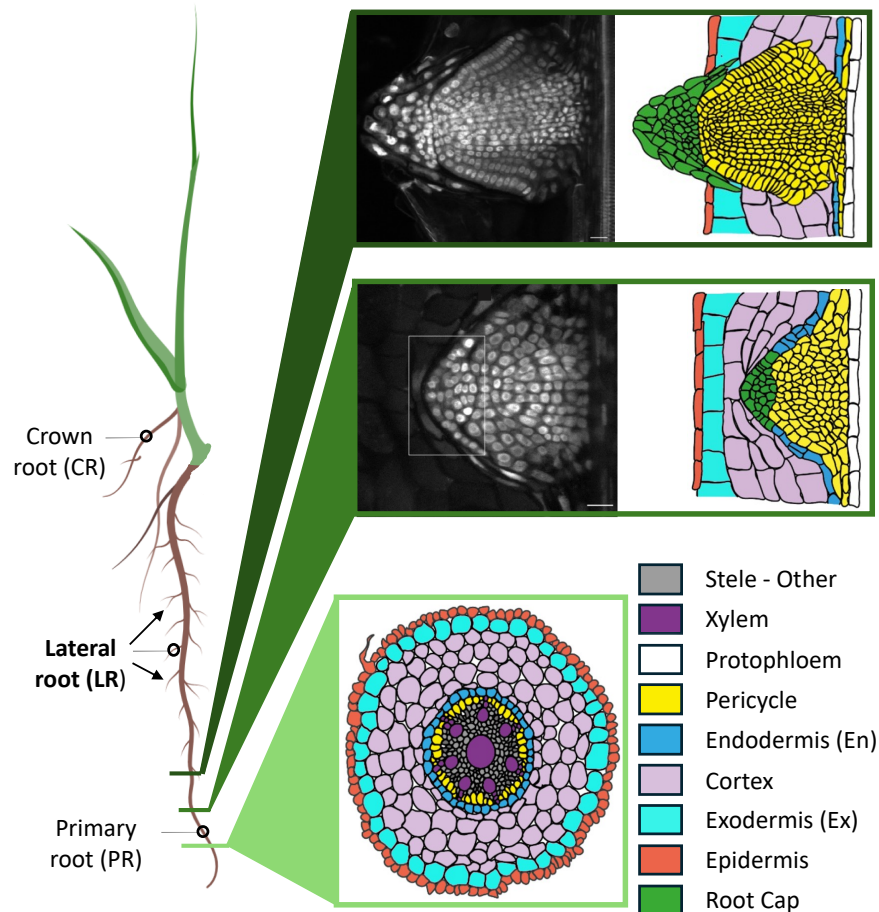




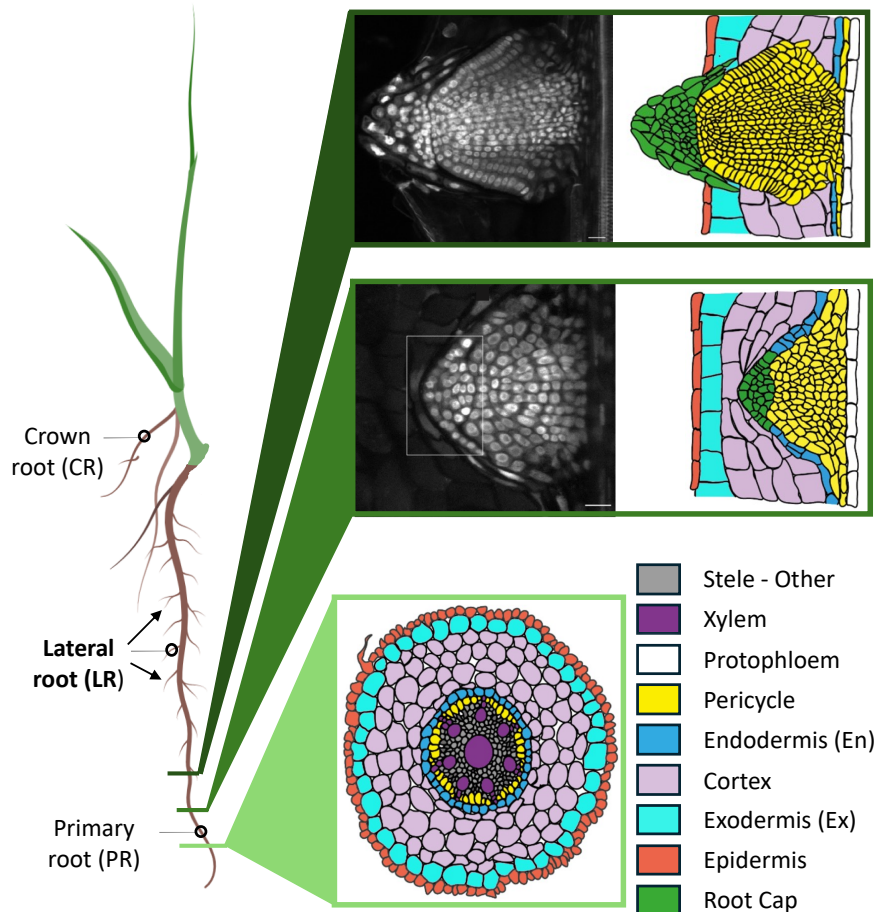


- Epidermis
- Cortex
- Endodermis
- Pericycle
- Xylem
- Phloem
- Stele

How does a new root emerge in monocotyledons with an exodermis?



How does a new root emerge in monocotyledons with an exodermis?



- Where do new roots come from?
- Which cells decide to become a new root?
- How does a new root grow through existing tissues?
- How does the environment influence this process?
- Can roots change their architecture to explore the soil?

New tools to explore the hidden world of roots

What is single cell sequencing ?



GTATGTGCAACG
GCAACGTGCCCA
TGCCCAAGTGGC



Bulk RNAseq

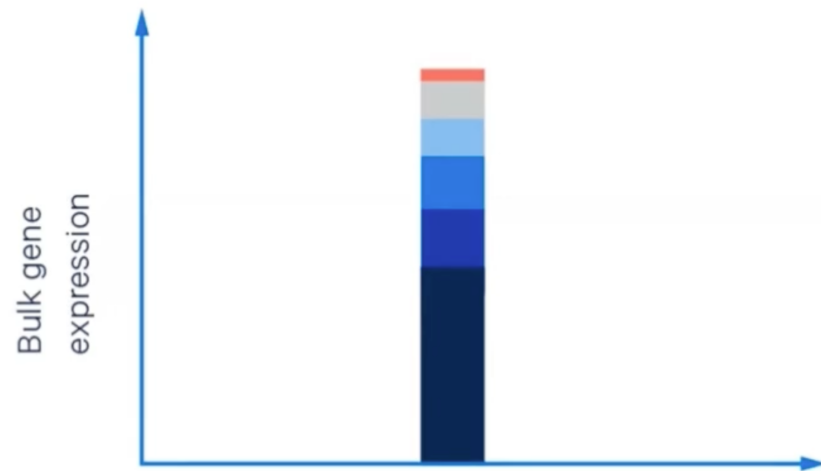


Single cell
resolution

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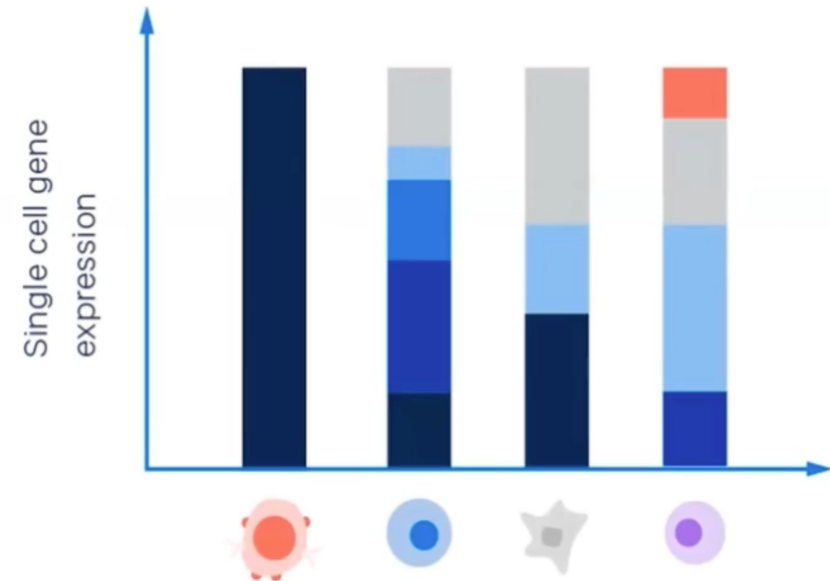
What is single cell sequencing ?

Bulk RNA-sequencing



Average gene expression from all cells

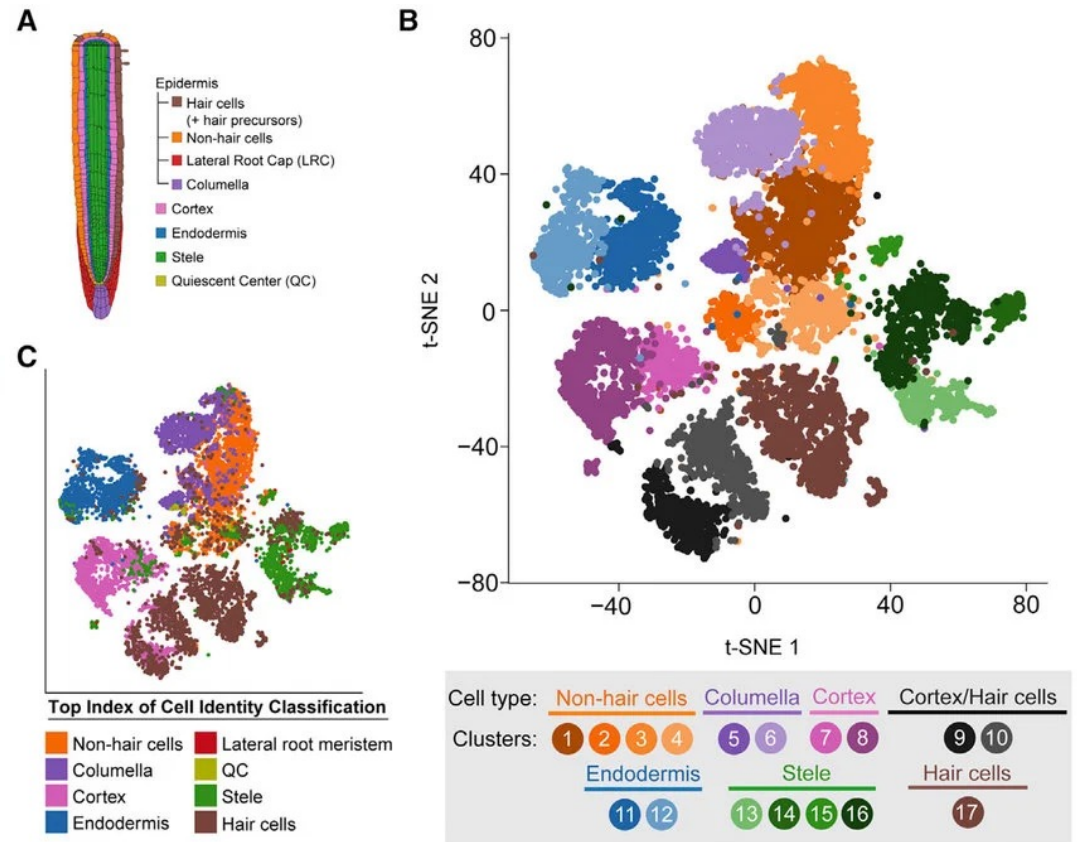
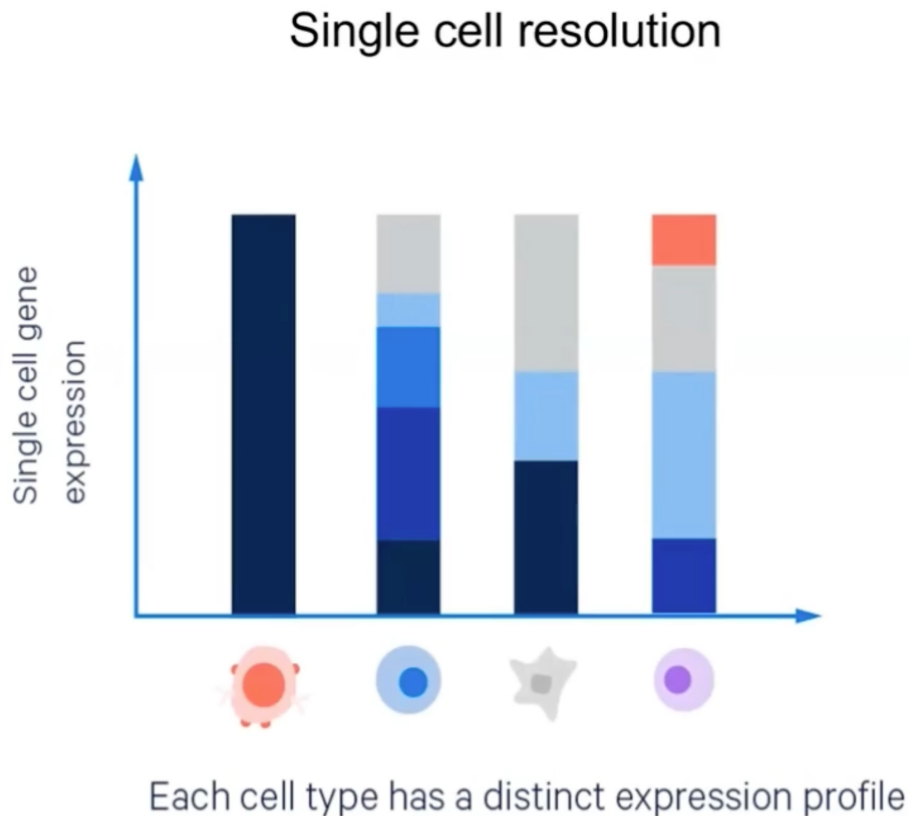
Single cell resolution



Each cell type has a distinct expression profile

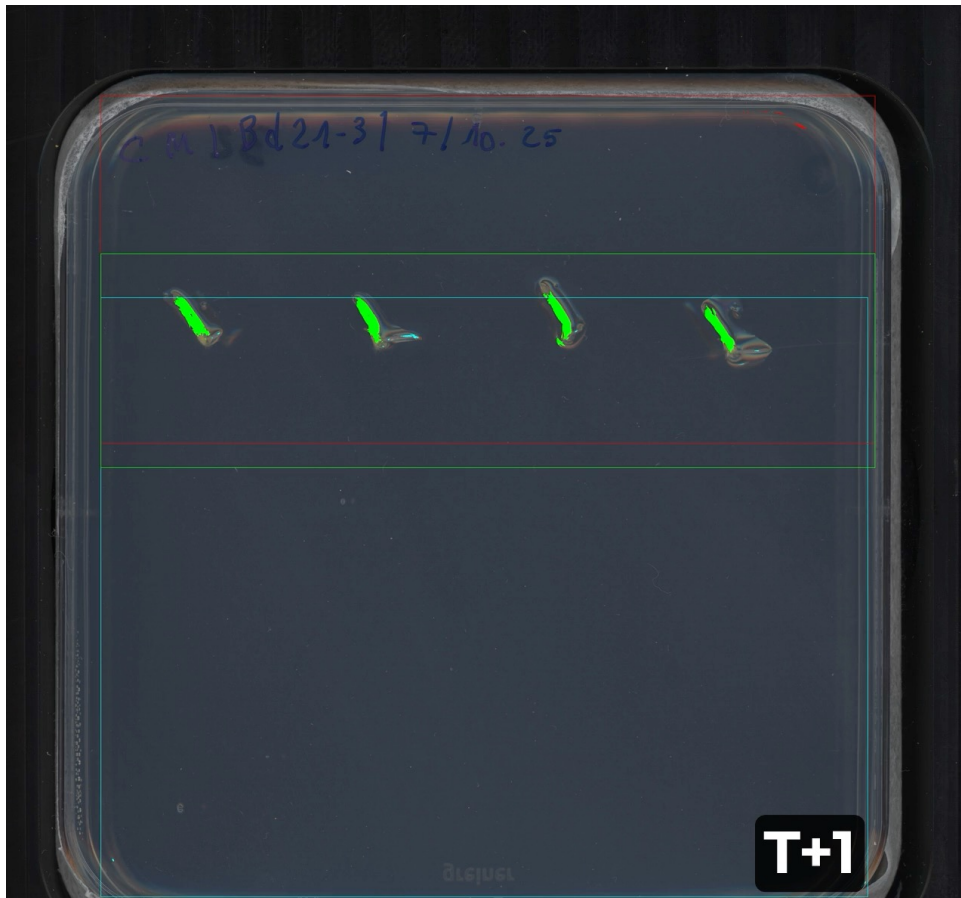
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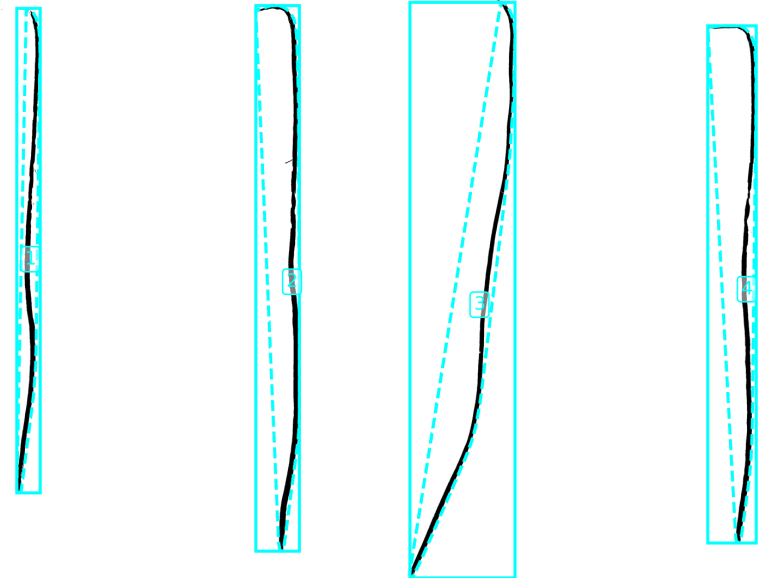


New tools to explore the hidden world of roots

AI-assisted root phenotyping

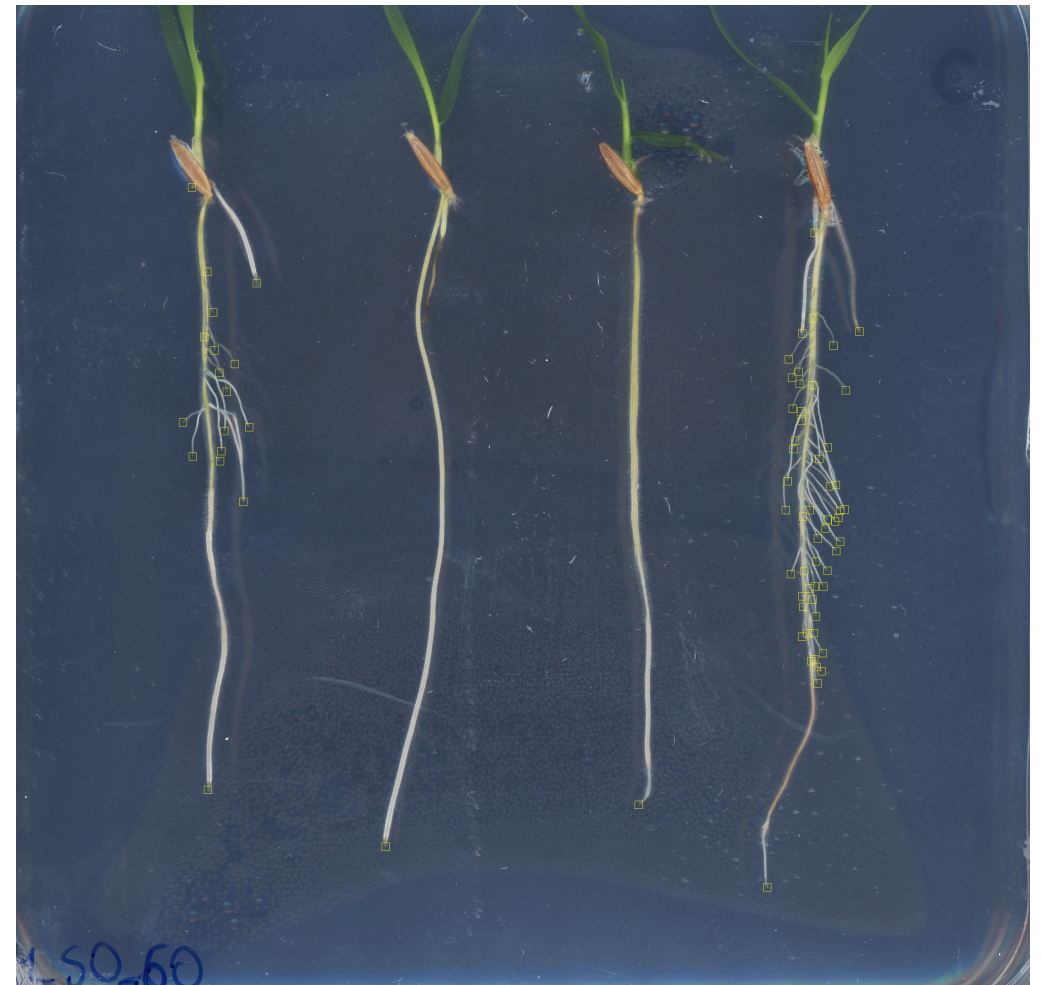
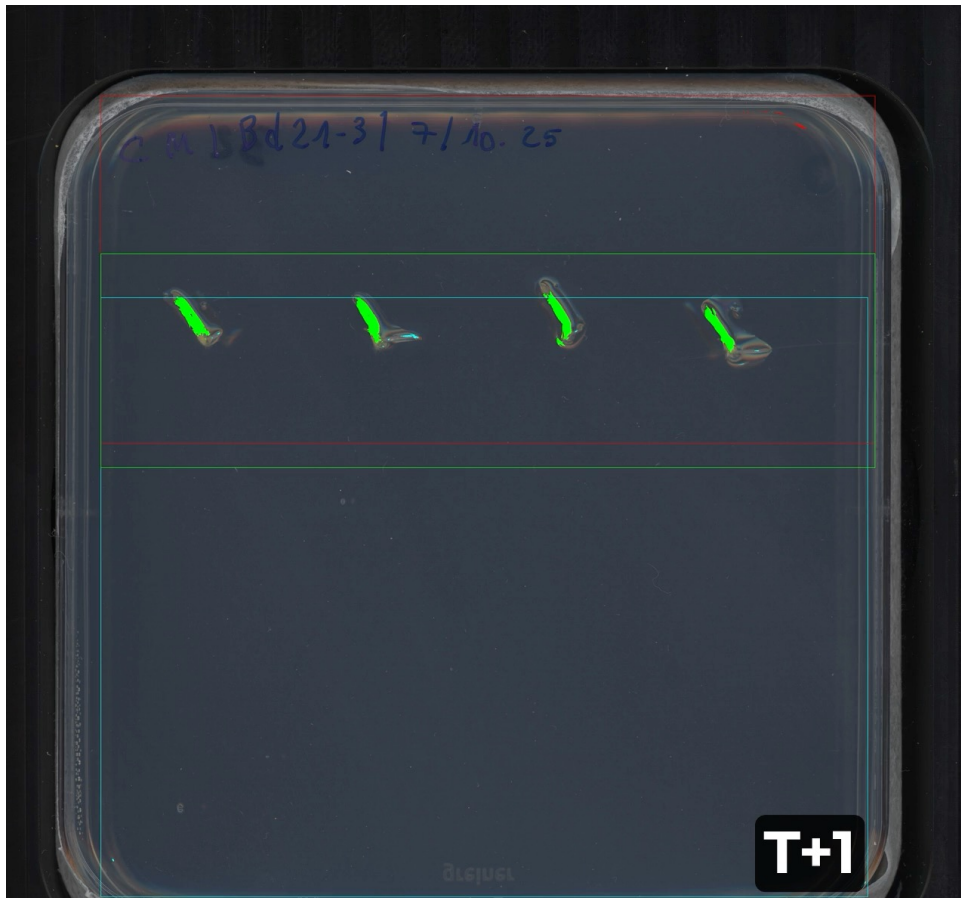


Bd_20251020_032 - root (min_area=1500, close=5, dilate=2, tol=15, holes $\leq$ 200, rm_small $\geq$ 0, merge_



New tools to explore the hidden world of roots

AI-assisted root phenotyping



Summary

- Studying plants can help find solutions to major issues facing the planet
- Roots play important roles in plant growth and health
- Not a passive tubes
- Sense soil signals
- Adjust growth and branching
- Control what enters in the plant